RAMCO INSTITUTE OF TECHNOLOGY

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NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Mechanical Engineering

Academic Year 2023 – 2024 (Odd Semester)

Name of the Faculty	:	Dr. J.Jerold John Britto AP(Sr.Gr.)/Mech
Course Code and Title	:	CME340 CAD/CAM (Honors Course)
Year and Branch	:	III -Mechanical
Unit/ Topic	:	II – 2D Drafting
Course Outcome	:	Develop the two-dimensional drafting and projection views
Topic Learning Outcome	:	Creation of 2D drawings including title blocks, BOMs, notes, and balloon annotations to facilitate efficient inspection reporting.
Activity Chosen		Hands-On Workshop (Industry Assisted)
Time Allotted for Activity	:	2 days
Date	:	30.08.2023 & 31.08.2023
No.of Students participated	:	10
Justification	:	The chosen topic for industry-assisted software training, focused on 2D drafting with a specific emphasis on Title Block creation, Bill of Materials (BOM) creation, notes incorporation, and the ballooning of 2D drawings within PTC Creo 8.0, holds significant relevance and practicality. In industries heavily reliant on precise engineering documentation, such as manufacturing and design, mastering these features is crucial for creating comprehensive and standardized drawings. The Title Block establishes essential information about the drawing, BOM creation streamlines the manufacturing process, notes contribute to effective communication, and ballooning aids in inspection reporting by associating callouts

		with specific parts or components. This training equips professionals with the essential skills to enhance efficiency, accuracy, and communication in the design and manufacturing processes, aligning perfectly with industry demands and the capabilities of the PTC Creo 8.0 software.
Details of Implementation	:	**Day 1: August 30th, 2023 (Wednesday)** Morning Session (9:00 AM - 12:30 PM) 1. **Introduction to PTC CREO 8.0** • Overview of the software interface. • Introduction to 2D Drafting capabilities. 2. **Part Drawing in CREO Parametric** • Creating a basic part in CREO Parametric. • Understanding the part modeling features. 3. **Drafting Environment** • Transitioning from 3D to 2D. • Understanding the drafting environment. Afternoon Session (1:30 PM - 5:00 PM) 4. **Title Block Creation** • Importance of Title Blocks in engineering drawings. • Step-by-step guide to creating and customizing Title Blocks. 5. **BOM Creation** • Understanding the Bill of Materials. • Creating a BOM in a drawing. **Day 2: August 31st, 2023 (Thursday)** Morning Session (9:00 AM - 12:30 PM) 6. **Advanced Drafting Tools**

- Exploring advanced drafting tools in CREO.
- Tips and tricks for efficient drafting.

7. **Notes Creation**
   - Adding annotations and notes to drawings.
   - Using the text and annotation tools effectively.

Afternoon Session (1:30 PM - 5:00 PM)

8. **Ballooning of 2D Drawings**
   - Understanding the importance of ballooning for inspection.
   - Step-by-step guide to adding balloons to a drawing.
9. **Inspection Reporting**
   - Overview of inspection reporting in CREO.
   - Generating reports and exporting them.
10. **Q&A and Hands-On Practice**
    - Open floor for questions.
    - Hands-on practice on the topics covered.

Additional Information:

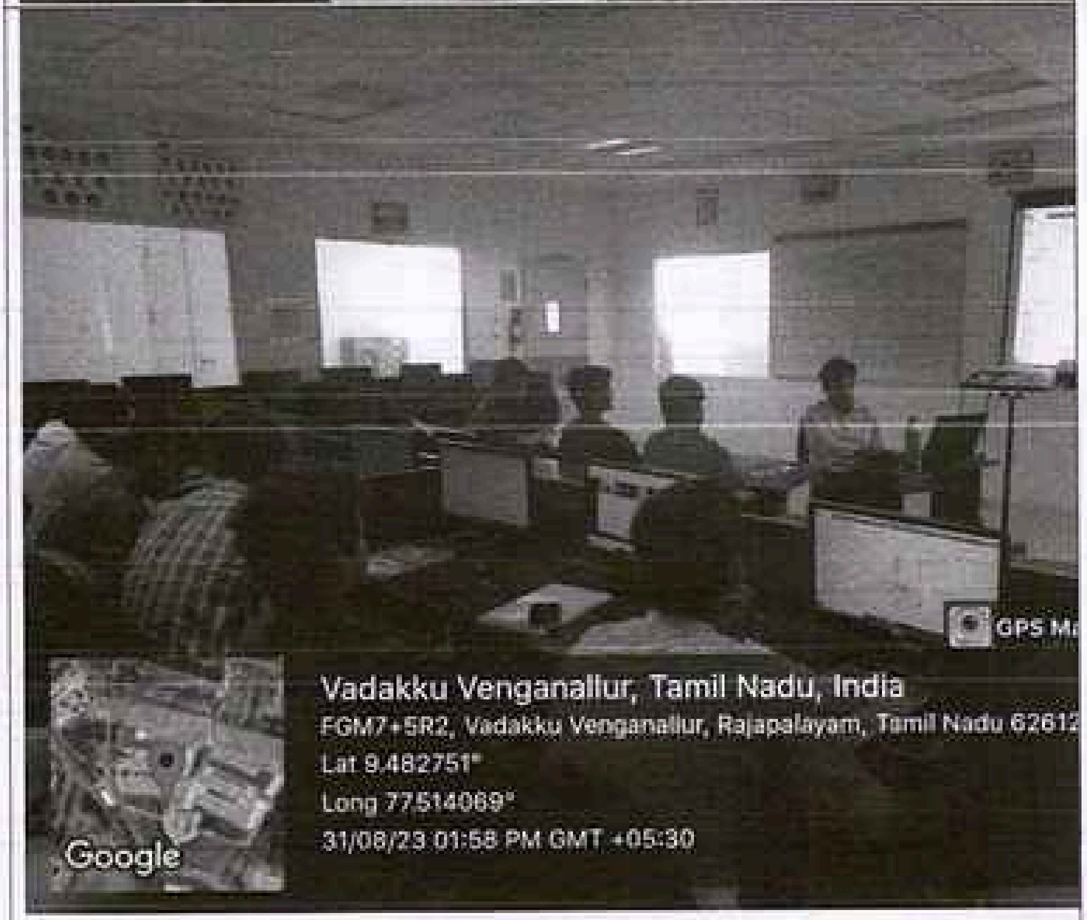
- **Facilitator Profile:**
  - Keerthi Vasan V, Application Engineer, Adroitec Engineering Solutions Pvt Ltd. Chennai.
- **Venue:**
  - RIT-Harita Centre of Excellence

PO/PSO mapping for the activity

Course Outcome /Programme Outcome	PO1	PO3	PO5	P12	PSO1
Level of Mapping	2	2	3	2	2
Justification for correlation	Demonstrate the application of engineering knowledge and skills in creating 2D drawings using PTC CREO 8.0 for part drawings, drafting, and projection views.	Develop solutions for 2D drafting challenges by utilizing the features of PTC CREO 8.0, including part drawing tools and drafting capabilities.	Use industry-standard software tools, specifically PTC CREO 8.0, for creating 2D drawings, title blocks, BOMs, notes, and ballooning in the context of inspection reporting.	Consider environmental and sustainability aspects in 2D drafting practices, promoting efficiency and resource optimization.	To design component using CAD package

(1 – Low, 2- Moderate, 3-High)

Activity Glimpses



Reflective Critique:

i. Feedback of practice from students and other stakeholders:

Students gave positive oral feedback

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Technical Call Report

Customer Name: RANCO INSTITUTE OF TECHNOLOGY.	**Call Date:** 20/8/23 & 21/8/23
Contact Person: Dr. JEROLD.	Time In / Time Out
Designation&Dept.: Asst prof. senior grade. MECH.	Duration:
Address: RAJAPALAYAM.	Call Type/Nature of Work: Demo ( ) Installation( ) Training(✓) Support( ) AMC Visit( ) Others:
Phone: 8427826827. Extension:	Fax:
E-Mail: jerold@ritrjpm.ac.in.	

Software Details

Name of the Software: CREO 9.0.

Account Type: Commercial / Education

Total No. Of License:

System/Network Details

Operating System: Service Pack: Network Type: Windows( ) Novell( )
System Config: Ram:
Graphics Card Details:

Installation Details

Installation Type: Standalone ( ) Network ( )

If Standalone Specify

If Network, Specify **Server code/Host ID:**

For IronCAD:
Serial Number:
Code Word:
Machine Lock ID:
Comments:

Support Call Details

Objective of Visit:	**Solution Offered:**
CREO TRAINING.	→ TRAINING ON:- Sketcher, Partmodeling, Surfacing, Assembly & Detailing. → 10 students Attended.
Customer's Remarks/Feedback: [signature] 21/8/23 **Customer's Signature & Seal** Department of Mechanical Engineering Ramco Institute of Technology Rajapalayam - 626 117.	**Executive Remarks/Feedback:** Signature: [signature] Executive Name: KEERTHI VASAN VIJAYAKUMAR
	Technical Manager/RSM -Remarks: Manager Signature

NAME	CONTACT & MAIL ID.	FEED BACK.
1) BALAJI S.	9566913497 95362111 4004 @ nitrjpm.ac.in	I am very happy to interact with you, clearing many of my doubts, gained knowledge about new software
2) GANESH KUMAR P	6381188792 95362111 4010 @ nitrjpm.ac.in	very helpful session and also very interactive and we have learnt a new software CREO which will be useful
3) GOWTHAM J	8148023112 95362111 4012 @ nitrjpm.ac.in	It is very useful session and we learnt a good designing software which will be useful for us.
4) HARISH KRISHNAN R	8248414005 953621114014 @ nitrjpm.ac.in	It is very useful to improve the knowledge about the creo software. Majorly all details of the software are known.
5) JEYA PRAKASH K	9585189142 jpjayaprakash012 @gmail.com	The session was very interactive and helpful. Gained knowledge about CREO software.
6) GURUSANKAR N	9952607079 n.gurusankar@gmail.com	the session was very helpful to gained more knowledge about CREO Software.
7) RAM KUMAR U	7358093621 95362111 4036 @nitrjpm.ac.in	The session was very informative. Interaction is very good.
8) SUDARSHAN R.	9489067430 sudarshanramamoorthi@gmail.com	Gained knowledge about CREO software which was new to us.
9) MUTHUKUMAR. R.	9025950145 muthukumar 173439@gmail.com	Gained knowledge about CREO Software and Basic Details about Industry drawing which was new
10) VARUN .K.	8148055456 95362111 4056 @nitrjpm.ac.in	—

ii. Benefit of the practice:

Students learnt to use CAD software package and file format topics.

iii. Whether the practice is adopted in any of the courses early:

Yes, in the same course

Challenges faced in implementation:

Balancing comprehensive 2D drafting, including title block creation, BOM, notes, and ballooning, within a two-day PTC CREO 8.0 training, with potential challenges in ensuring thorough understanding and application due to time constraints.

References: Nil

CO2: Students will be able to Develop the two-dimensional drafting and projection views.

Faculty In Charge **HoD/Mech**

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Department of Mechanical Engineering

Academic Year 2022 – 2023 (Even Semester)

Name of the Faculty	:	Dr. J.Jerold John Britto AP(Sr.Gr.)/Mech
Course Code and Title	:	GE3251 Engineering Graphics
Year and Branch	:	I -Mechanical
Unit/ Topic	:	Unit-III Projection of Solids
Course Outcome	:	Construct the projection of simple solids and free hand sketches of multiple views from pictorial views of objects.
Topic Learning Outcome	:	Demonstrate proficiency in generating engineering graphics by constructing projections of simple solids and producing free-hand sketches of multiple views from pictorial views, including hands-on fabrication of solid models using card sheets for prisms and pyramids.
Activity Chosen		Hands-On Solid Modelling: From Pictorial Views to Card Sheet Creations
Time Allotted for Activity	:	50 minutes
Date	:	28.06.2023
No.of Students participated	:	36
Justification	:	The fabrication of simple solid models using card sheet for the projection of solids in the context of the GE3251 Engineering Graphics course serves as a hands-on and practical application of theoretical knowledge. This activity enhances students' understanding of projection principles by challenging them to translate pictorial views of objects into accurate, scaled representations. Constructing solid models such as prisms and pyramids using card sheets

		not only reinforces their comprehension of the unit on Projection of Solids but also hones their spatial visualization skills. Additionally, the free-hand sketches of multiple views derived from these fabricated models further cultivate their ability to communicate design ideas through engineering drawings. Overall, this activity bridges the gap between theory and application, fostering a holistic learning experience in mechanical engineering graphics.
Details of Implementation	:	The implementation of the "Projection of Solids" activity in the GE3251 Engineering Graphics course involved a hands-on approach with a total time duration of 120 minutes. The activity focused on fabricating simple solid models using card sheets, specifically prisms and pyramids. Here are the details of the implementation: **Introduction (5 minutes):** The session began with a brief overview of the importance of projection in engineering graphics, emphasizing its role in communication and design. The specific topic of "Projection of Solids" was introduced, highlighting its relevance in understanding three-dimensional objects. **Theory and Demonstration (20 minutes):** A short lecture was conducted on the principles of projecting solids, explaining orthographic projection and its significance in representing objects accurately. Examples were provided to illustrate how simple solids like prisms and pyramids could be projected onto a two-dimensional plane. The process was demonstrated using visual aids and illustrations. **Hands-on Fabrication (70 minutes):** The majority of the time was allocated for the practical aspect. Card sheets, rulers, pencils, and other necessary materials

		were distributed. Students were instructed to fabricate simple solid models, focusing on prisms and pyramids. Creativity and precision were encouraged, and guidance was provided by circulating among students. **Sketching Multiple Views (20 minutes):** Once the fabricated models were complete, students were guided in creating free-hand sketches of multiple views based on their solid models. Emphasis was placed on maintaining proportion and accuracy in representing different views (top, front, and side). Common challenges and solutions were discussed during this sketching process. **Review and Discussion (5 minutes):** The activity concluded with a short review of the key concepts covered. A brief discussion was facilitated on the challenges encountered during fabrication and sketching. Students were encouraged to share their experiences and insights. By combining theory, hands-on fabrication, and sketching, this implementation ensured a well-rounded understanding of the projection of solids in engineering graphics

PO/PSO mapping for the activity

Course Outcome /Programme Outcome	PO1	PO3	PO5	P12	PSO1
Level of Mapping	2	2	3	2	2
Justification for correlation	Understand the principles of CNC machining	Develop solutions for CNC machining tasks	Utilize computer-aided machining tools for	Recognize the need for continuous learning in	To design component using CAD package

	and part programming for MTAB Flexturn & FlexMill.	using MTAB Flexturn & FlexMill.	writing and optimizing CNC part programs.	the rapidly evolving field of CNC machining.	

(1 – Low, 2- Moderate, 3-High)

Activity Glimpses

Reflective Critique:

i. Feedback of practice from students and other stakeholders:

Students gave positive oral feedback

ii. Benefit of the practice:

Practicing the fabrication of simple solid models using card sheets, such as prisms and pyramids, in the context of the "Projection of Solids" unit in Engineering Graphics, enhances spatial visualization skills and reinforces the correlation between 3D objects and their 2D projections, fostering a deeper understanding of technical drawing principles.

iii. Whether the practice is adopted in any of the courses early:

Yes, in the same course

Challenges faced in implementation:

Implementing the activity of fabricating simple solid models using card sheets, such as prisms and pyramids, for the topic "Projection of Solids" in the GE3251 Engineering Graphics class poses several challenges. Firstly, coordinating the construction of these models within the allotted 120-minute time frame can be demanding, requiring efficient time management. Secondly, ensuring that students accurately translate pictorial views into free-hand sketches of multiple views demands a strong grasp of engineering graphics principles. Additionally, challenges may arise in providing personalized guidance to students, addressing their individual struggles in fabricating and sketching the solids. Balancing the hands-on fabrication with the theoretical understanding of projections may be challenging, underscoring the need for a well-structured and engaging instructional approach.

References: Nil

CO3: Students will be able to Construct the projection of simple solids and free hand sketches of multiple views from pictorial views of objects.

Faculty In Charge

HoD/Mech